

DSA0602 Data Handling and Visualization

Unit II - Class Activity 1 Solutions & Visual Evaluation Report

Date: 20.07.2026 | Level: K5 Evaluation

Scenario 1: Hospital Patient Admission Analysis

Scenario Description: A hospital collected data on patient admissions across multiple departments (Cardiology, Neurology, Pediatrics, Orthopedics, Emergency) for the past year. Evaluate which visualization effectively communicates department workload, admission variability, and patient distribution.

Suggested Libraries: Python: Pandas, Matplotlib, Seaborn, Plotly | R: ggplot2, dplyr

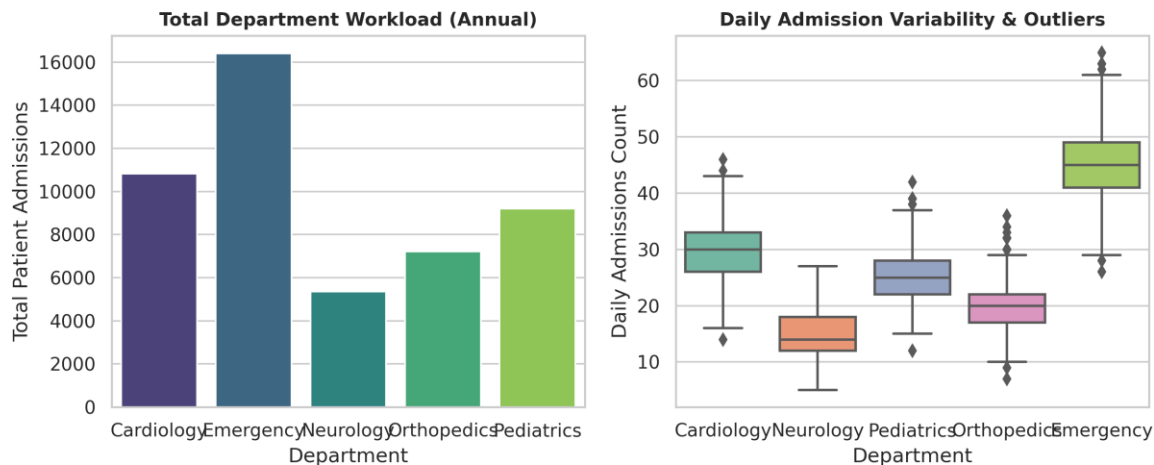
Python Implementation Code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Generate Synthetic Data
departments = ["Cardiology", "Neurology", "Pediatrics", "Orthopedics", "Emergency"]
dates = pd.date_range(start="2025-01-01", periods=365, freq="D")
data = []
for d in dates:
    for dept in departments:
        lam_val = 45 if dept == "Emergency" else (30 if dept == "Cardiology" else 20)
        data.append({"Date": d, "Department": dept, "Admissions":
np.random.poisson(lam=lam_val)})
df = pd.DataFrame(data)

# Visualization
fig, axes = plt.subplots(1, 2, figsize=(10, 4.2))
total_adm = df.groupby("Department")["Admissions"].sum().reset_index()
sns.barplot(data=total_adm, x="Department", y="Admissions", ax=axes[0], palette="viridis")
sns.boxplot(data=df, x="Department", y="Admissions", ax=axes[1], palette="Set2")
plt.tight_layout()
plt.show()
```

Generated Visual Output Dashboard:



Evaluation & Justification (K5 Level Analysis):

- **Department Workload:** The Bar Chart effectively communicates total patient volume per department, showing high absolute capacity requirements in Emergency and Cardiology.
- **Admission Variability:** The Box Plot is superior for identifying daily admission distribution spread, median volume, and unpredictable peak demand days (outliers).
- **Executive Recommendation:** Management should use dynamic dashboard Box Plots for daily staffing allocation while using Bar Charts for annual budget planning.

Scenario 2: E-Commerce Customer Spending Behaviour

Scenario Description: An online retailer has customer purchase amounts, order frequency, and product categories. Create multiple visualizations to identify high-value customers, spending variability, and purchasing trends. Recommend the best visualization for business executives.

Suggested Libraries: *Python: Seaborn, Plotly Express / R: ggplot2*

Python Implementation Code:

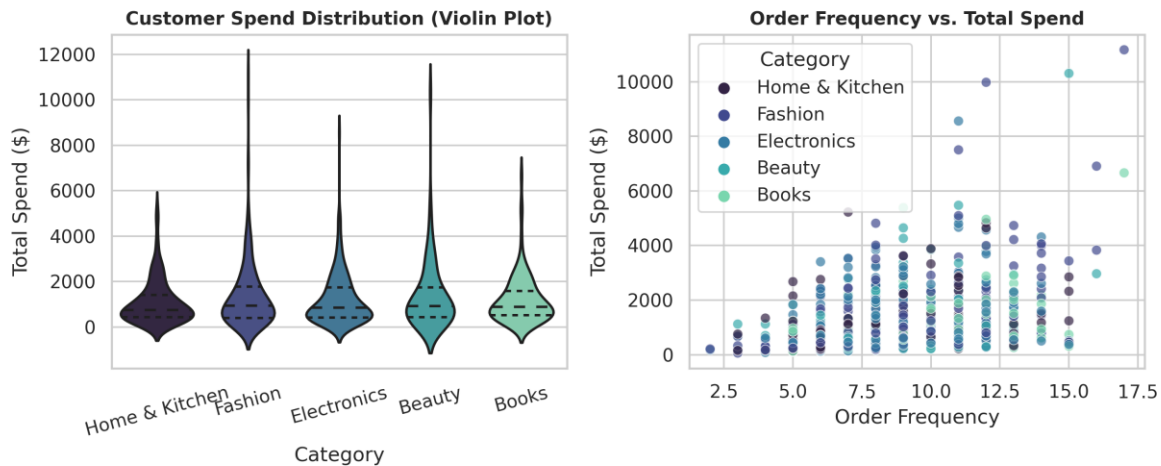
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

categories = ["Electronics", "Fashion", "Home & Kitchen", "Books", "Beauty"]
df = pd.DataFrame({
    "Category": np.random.choice(categories, 800),
    "OrderFrequency": np.random.poisson(lam=8, size=800) + 1,
    "AvgPurchaseAmount": np.random.exponential(scale=120, size=800) + 20
})
df["TotalSpend"] = df["OrderFrequency"] * df["AvgPurchaseAmount"]

fig, axes = plt.subplots(1, 2, figsize=(10, 4.2))
sns.violinplot(data=df, x="Category", y="TotalSpend", palette="mako", inner="quartile",
ax=axes[0])
sns.scatterplot(data=df, x="OrderFrequency", y="TotalSpend", hue="Category", palette="mako",
```

```
ax=axes[1])
plt.tight_layout()
plt.show()
```

Generated Visual Output Dashboard:



Evaluation & Justification (K5 Level Analysis):

- **High-Value Customer Identification:** The Violin Plot clearly displays spending density along with upper-quartile VIP customers across categories.
- **Purchasing Trends:** Scatter plots combining Order Frequency vs Total Spend reveal strong positive correlation clusters among high-frequency buyers.
- **Executive Recommendation:** Violin plots are recommended for executive dashboards as they capture both distribution shape and extreme spending outliers in a single view.

Scenario 3: University Examination Performance Dashboard

Scenario Description: A university wants to compare student marks across departments and identify score distributions, outliers, and top-performing classes. Evaluate chart options for academic decision-making.

Suggested Libraries: *Python: Pandas, Seaborn, Plotly / R: ggplot2, plotly*

Python Implementation Code:

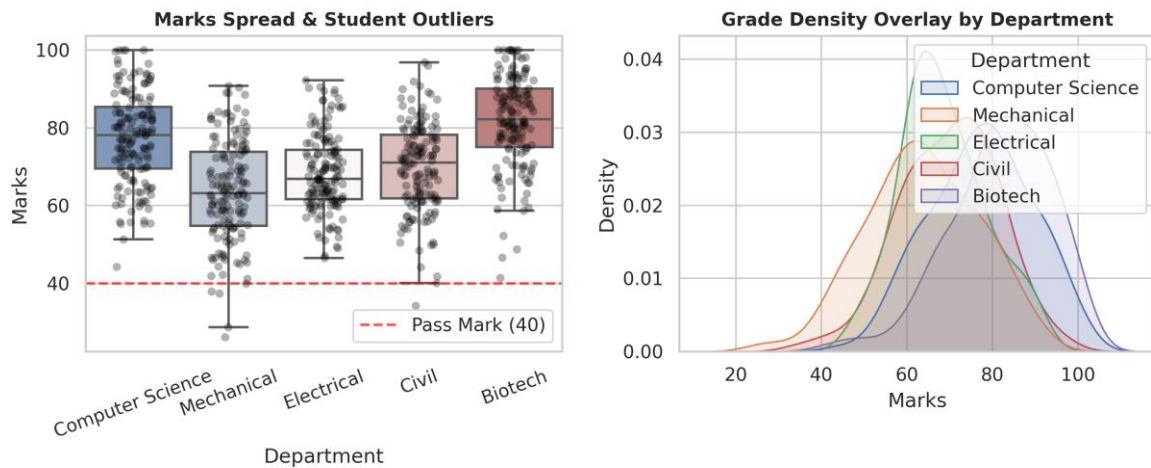
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

depts = ["Computer Science", "Mechanical", "Electrical", "Civil", "Biotech"]
records = []
for d in depts:
    scores = np.clip(np.random.normal(70, 12, 150), 0, 100)
    for s in scores:
        records.append({"Department": d, "Marks": s})
```

```
df = pd.DataFrame(records)
```

```
fig, axes = plt.subplots(1, 2, figsize=(10, 4.2))
sns.boxplot(data=df, x="Department", y="Marks", ax=axes[0], palette="vlag", fliersize=0)
sns.stripplot(data=df, x="Department", y="Marks", ax=axes[0], color="black", alpha=0.3)
axes[0].axhline(40, color="red", linestyle="--")
sns.kdeplot(data=df, x="Marks", hue="Department", fill=True, alpha=0.15, ax=axes[1])
plt.tight_layout()
plt.show()
```

Generated Visual Output Dashboard:



Evaluation & Justification (K5 Level Analysis):

- **Score Distribution & Outliers:** Box Plots overlaid with Strip Plots (raw student points) precisely locate failing outliers and top-performing students.
- **Department Comparisons:** Density Overlay (KDE) curves illustrate comparative grade distributions (e.g., Computer Science high mean vs Mechanical variance).
- **Dashboard Improvements:** Include target policy reference lines (e.g., pass mark = 40, distinction = 75) directly on charts for actionable interventions.

Scenario 4: Climate and Rainfall Distribution Study

Scenario Description: Meteorological data contains monthly rainfall, temperature, and humidity for multiple cities. Evaluate which visualization best represents seasonal rainfall variation and climate distribution for researchers.

Suggested Libraries: Python: Matplotlib, Seaborn | R: ggplot2

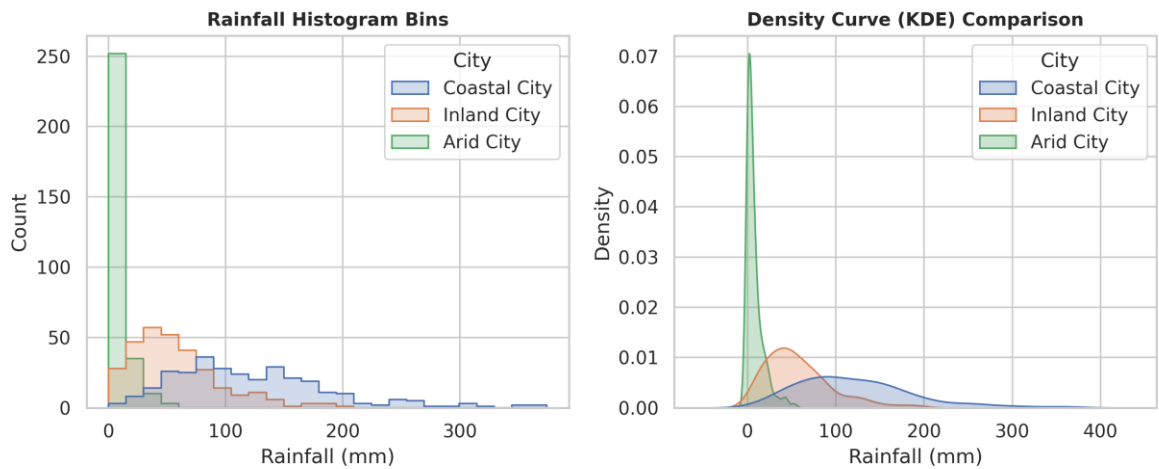
Python Implementation Code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
cities = ["Coastal City", "Inland City", "Arid City"]
records = []
for city in cities:
    shape_val = 3.0 if "Coastal" in city else (0.5 if "Arid" in city else 2.0)
    rf = np.random.gamma(shape=shape_val, scale=30, size=300)
    for val in rf:
        records.append({"City": city, "Rainfall_mm": val})
df = pd.DataFrame(records)

fig, axes = plt.subplots(1, 2, figsize=(10, 4.2))
sns.histplot(data=df, x="Rainfall_mm", hue="City", element="step", ax=axes[0])
sns.kdeplot(data=df, x="Rainfall_mm", hue="City", fill=True, alpha=0.3, ax=axes[1])
plt.tight_layout()
plt.show()
```

Generated Visual Output Dashboard:



Evaluation & Justification (K5 Level Analysis):

- **Continuous Data Representation:** Rainfall data is naturally right-skewed. Continuous Density Plots (KDE) offer smoother probabilistic representations compared to discrete Histograms.
- **City Comparisons:** Overlaid KDE curves allow instant visual comparison of climate severity across Coastal, Inland, and Arid regions.
- **Researcher Justification:** KDE curves avoid bin-width bias inherent in histograms, making them the most scientifically rigorous tool for environmental papers.

Scenario 5: Bank Loan Risk Assessment

Scenario Description: A financial institution has loan amounts, repayment history, income levels, and credit scores. Evaluate visualizations showing loan distributions and customer segments supporting approval decisions.

Suggested Libraries: *Python: Seaborn, Plotly, Pandas / R: ggplot2*

Python Implementation Code:

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

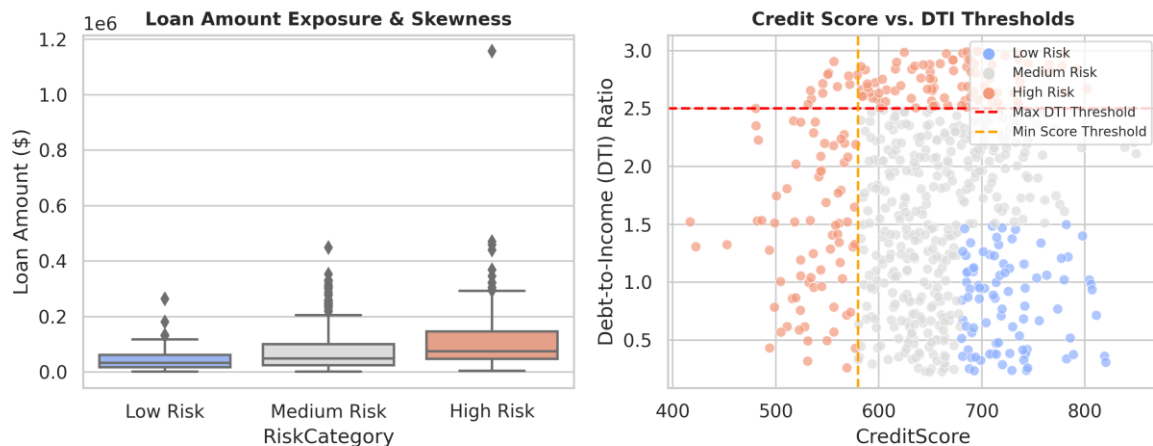
income = np.random.lognormal(mean=10.5, sigma=0.75, size=600)
loan_amount = income * np.random.uniform(0.2, 3.0, size=600)
credit_score = np.random.normal(650, 70, size=600).clip(300, 850)
dti = loan_amount / income

risk = ["High Risk" if s < 580 or r > 2.5 else ("Medium Risk" if s < 680 or r > 1.5 else "Low Risk") for s, r in zip(credit_score, dti)]
df = pd.DataFrame({"LoanAmount": loan_amount, "CreditScore": credit_score, "DTI": dti, "RiskCategory": risk})

fig, axes = plt.subplots(1, 2, figsize=(10, 4.2))
sns.boxplot(data=df, x="RiskCategory", y="LoanAmount", palette="coolwarm", ax=axes[0])
sns.scatterplot(data=df, x="CreditScore", y="DTI", hue="RiskCategory", palette="coolwarm", ax=axes[1])
axes[1].axhline(2.5, color="red", linestyle="--")
axes[1].axvline(580, color="orange", linestyle="--")
plt.tight_layout()
plt.show()

```

Generated Visual Output Dashboard:



Evaluation & Justification (K5 Level Analysis):

- **Risk Pattern Isolation:** Scatter Plots comparing Credit Score vs. Debt-To-Income (DTI) with threshold overlays delineate clear approval/rejection boundaries.
- **Skewness & Outlier Exposure:** Box Plots by Risk Category reveal extreme exposure limits in high-risk loan applicant requests.
- **Management Dashboard Recommendation:** Combine the threshold Scatter Plot with exposure Box Plots to empower credit risk officers with automated screening rules.